MyMusicTaste

0.1.0

Generated by Doxygen 1.14.0

1 Namespace Index 1

1.1 Namespace List . . . 1

2 Hierarchical Index 3

2.1 Class Hierarchy . . . 3

3 Class Index 5

3.1 Class List . . . 5

4 Namespace Documentation 9

4.1 MyMusicTaste Namespace Reference . . . 9

4.2 MyMusicTaste.Components Namespace Reference . . . 9

4.3 MyMusicTaste.Components.Comments Namespace Reference . . . 9

4.4 MyMusicTaste.Components.Dialogs Namespace Reference . . . 9

4.5 MyMusicTaste.Components.Misc Namespace Reference . . . 10

4.6 MyMusicTaste.Components.Page_Auth Namespace Reference . . . 10

4.7 MyMusicTaste.Components.Page_Home Namespace Reference . . . 10

4.8 MyMusicTaste.Components.Page_Search Namespace Reference . . . 10

4.9 MyMusicTaste.Components.Page_Song Namespace Reference . . . 10

4.10 MyMusicTaste.Components.Page_SongSubmission Namespace Reference . . . 11

4.11 MyMusicTaste.Components.Page_User Namespace Reference . . . 11

4.12 MyMusicTaste.Database Namespace Reference . . . 11

4.13 MyMusicTaste.Database.Contexts Namespace Reference . . . 11

4.14 MyMusicTaste.Database.Contexts.MongoDb Namespace Reference . . . 11

4.15 MyMusicTaste.Database.Contexts.MongoDb.Operations Namespace Reference . . . 12

4.16 MyMusicTaste.Database.Operations Namespace Reference . . . 12

4.17 MyMusicTaste.Models Namespace Reference . . . 13

4.18 MyMusicTaste.Startup Namespace Reference . . . 13

4.19 MyMusicTaste.Utils Namespace Reference . . . 13

4.19.1 Function Documentation . . . 13

4.19.1.1 CssSize() . . . 13

5 Class Documentation 15

5.1 AuthController Class Reference . . . 15

5.1.1 Member Function Documentation . . . 15

5.1.1.1 LoginAsync() . . . 15

5.1.1.2 LogoutAsync() . . . 16

5.2 Comment Class Reference . . . 16

5.3 CommentComp Class Reference . . . 17

5.4 CommentSection Class Reference . . . 17

5.5 ConfirmDialog Class Reference . . . 18

5.6 DatabaseOperationException Class Reference . . . 19

5.7 Dialog< TResult > Class Template Reference . . . 20

5.7.1 Detailed Description . . . 21

5.7.2 Member Function Documentation . . . 21
5.7.2.1 OpenDialog() . . . 21
5.8 DialogBase Class Reference . . . 21
5.9 DialogResult< TResult > Struct Template Reference . . . 22
5.9.1 Detailed Description . . . 22
5.10 Dropzone Class Reference . . . 22
5.10.1 Member Function Documentation . . . 23
5.10.1.1 SetActive() . . . 23
5.11 Histogram< TValue, TLabel > Class Template Reference . . . 23
5.11.1 Detailed Description . . . 23
5.12 ICommentsListing Interface Reference . . . 24
5.12.1 Member Function Documentation . . . 24
5.12.1.1 GetCommentsByPageAsync() . . . 24
5.12.1.2 GetCommentsByUserAsync() . . . 25
5.13 IDbRepository< TModel > Interface Template Reference . . . 25
5.13.1 Detailed Description . . . 25
5.13.2 Member Function Documentation . . . 26
5.13.2.1 CreateAsync() . . . 26
5.13.2.2 DeleteAsync() . . . 26
5.13.2.3 GetById() [1/2] . . . 26
5.13.2.4 GetById() [2/2] . . . 26
5.13.2.5 GetByIdAsync() . . . 27
5.13.2.6 GetByIdsAsync() . . . 27
5.13.2.7 UpdateAsync() . . . 27
5.14 IIdentityProvider Interface Reference . . . 28
5.14.1 Member Function Documentation . . . 28
5.14.1.1 AssignRoleAsync() . . . 28
5.14.1.2 AuthorizeUserById() . . . 29
5.14.1.3 GetUserId() . . . 29
5.14.1.4 IsAuthenticated() . . . 29
5.14.1.5 LoginUserAsync() . . . 29
5.14.1.6 LogOutUserAsync() . . . 30
5.14.1.7 SignUpUserAsync() . . . 30
5.15 InputDialog Class Reference . . . 30
5.16 ISearchOperation< TModel > Interface Template Reference . . . 32
5.16.1 Detailed Description . . . 32
5.16.2 Member Function Documentation . . . 32
5.16.2.1 SearchAsync() . . . 32
5.17 ISongRatingListing Interface Reference . . . 33
5.17.1 Member Function Documentation . . . 33
5.17.1.1 GetRatingsBySongAsync() . . . 33
5.17.1.2 GetRatingsByUserAsync() . . . 33

5.17.1.3 GetSongRatingAsync() . . . 34
5.18 ISongStatsCalculation Interface Reference . . . 34
5.18.1 Member Function Documentation . . . 34
5.18.1.1 CalculateSongStatsAsync() . . . 34
5.19 ISongSubmission Interface Reference . . . 35
5.19.1 Member Function Documentation . . . 35
5.19.1.1 SubmitSongAsync() . . . 35
5.20 IUserLoginDto Interface Reference . . . 36
5.21 IUserSignupDto Interface Reference . . . 36
5.22 IUserStatsCalculation Interface Reference . . . 37
5.22.1 Member Function Documentation . . . 37
5.22.1.1 CalculateUserStatsAsync() . . . 37
5.23 AuthController.LoginDto Class Reference . . . 37
5.23.1 Property Documentation . . . 38
5.23.1.1 Password . . . 38
5.23.1.2 Username . . . 38
5.24 LoginPage Class Reference . . . 38
5.25 Model Class Reference . . . 39
5.26 MongoCommentsListing Class Reference . . . 39
5.26.1 Member Function Documentation . . . 39
5.26.1.1 GetCommentsByPageAsync() . . . 39
5.26.1.2 GetCommentsByUserAsync() . . . 40
5.27 MongoDbContext Class Reference . . . 40
5.27.1 Member Function Documentation . . . 40
5.27.1.1 Connect() . . . 40
5.28 MongoIdentity Class Reference . . . 41
5.28.1 Constructor & Destructor Documentation . . . 42
5.28.1.1 MongoIdentity() . . . 42
5.28.2 Member Function Documentation . . . 42
5.28.2.1 AssignRoleAsync() . . . 42
5.28.2.2 AuthorizeUserById() . . . 42
5.28.2.3 Configure() . . . 43
5.28.2.4 GetUserId() . . . 43
5.28.2.5 IsAuthenticated() . . . 43
5.28.2.6 LoginUserAsync() . . . 43
5.28.2.7 LogOutUserAsync() . . . 44
5.28.2.8 SignUpUserAsync() . . . 44
5.29 MongoRepository< TModel > Class Template Reference . . . 44
5.29.1 Detailed Description . . . 45
5.29.2 Member Function Documentation . . . 45
5.29.2.1 CreateAsync() . . . 45
5.29.2.2 DeleteAsync() . . . 45

5.29.2.3 GetById() [1/2] . . . 46
5.29.2.4 GetById() [2/2] . . . 46
5.29.2.5 GetByIdAsync() . . . 46
5.29.2.6 GetByIdsAsync() . . . 47
5.29.2.7 UpdateAsync() . . . 47
5.30 MongoSongRatingListing Class Reference . . . 48
5.30.1 Member Function Documentation . . . 48
5.30.1.1 GetRatingsBySongAsync() . . . 48
5.30.1.2 GetRatingsByUserAsync() . . . 48
5.30.1.3 GetSongRatingAsync() . . . 49
5.31 MongoSongSearch Class Reference . . . 49
5.31.1 Member Function Documentation . . . 49
5.31.1.1 SearchAsync() . . . 49
5.32 MongoSongsStatsCalculation Class Reference . . . 50
5.32.1 Member Function Documentation . . . 50
5.32.1.1 CalculateSongStatsAsync() . . . 50
5.33 MongoUserSearch Class Reference . . . 51
5.33.1 Member Function Documentation . . . 51
5.33.1.1 SearchAsync() . . . 51
5.34 MongoUserStatsCalculation Class Reference . . . 51
5.34.1 Member Function Documentation . . . 52
5.34.1.1 CalculateUserStatsAsync() . . . 52
5.35 RateSongComp Class Reference . . . 52
5.36 SearchComponent< TModel > Class Template Reference . . . 53
5.36.1 Detailed Description . . . 53
5.37 SearchPage Class Reference . . . 53
5.38 Song Class Reference . . . 54
5.39 SongPage Class Reference . . . 54
5.40 SongRating Class Reference . . . 55
5.41 SongRatingItem Class Reference . . . 56
5.42 SongRatingSection Class Reference . . . 56
5.43 SongSearchItem Class Reference . . . 56
5.44 SongStats Class Reference . . . 57
5.45 SongSubmission Class Reference . . . 57
5.45.1 Member Function Documentation . . . 57
5.45.1.1 SubmitSongAsync() . . . 57
5.46 SongSubmissionPage Class Reference . . . 58
5.47 Thumbnail Class Reference . . . 58
5.48 User Class Reference . . . 59
5.49 UserPage Class Reference . . . 60
5.50 UserSearchItem Class Reference . . . 60
5.51 UserSignupForm Class Reference . . . 61

5.52 UserStats Class Reference 61

Index **63**

Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

MyMusicTaste 9
MyMusicTaste.Components 9
MyMusicTaste.Components.Comments 9
MyMusicTaste.Components.Dialogs 9
MyMusicTaste.Components.Misc 10
MyMusicTaste.Components.Page_Auth 10
MyMusicTaste.Components.Page_Home 10
MyMusicTaste.Components.Page_Search 10
MyMusicTaste.Components.Page_Song 10
MyMusicTaste.Components.Page_SongSubmission 11
MyMusicTaste.Components.Page_User 11
MyMusicTaste.Database 11
MyMusicTaste.Database.Contexts 11
MyMusicTaste.Database.Contexts.MongoDb 11
MyMusicTaste.Database.Contexts.MongoDb.Operations 12
MyMusicTaste.Database.Operations 12
MyMusicTaste.Models 13
MyMusicTaste.Startup 13
MyMusicTaste.Utils 13

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AuthController . . . 15
CommentComp . . . 17
CommentSection . . . 17
DatabaseOperationException . . . 19
Dialog< TResult > . . . 20
  ConfirmDialog . . . 18
  InputDialog . . . 30
DialogBase . . . 21
DialogResult< TResult > . . . 22
Dropzone . . . 22
Histogram< TValue, TLabel > . . . 23
ICommentsListing . . . 24
  MongoCommentsListing . . . 39
IDbRepository< TModel > . . . 25
  MongoRepository< TModel > . . . 44
IIdentityProvider . . . 28
  MongoIdentity . . . 41
ISearchOperation< TModel > . . . 32
  MongoSongSearch . . . 49
  MongoUserSearch . . . 51
ISongRatingListing . . . 33
  MongoSongRatingListing . . . 48
ISongStatsCalculation . . . 34
  MongoSongsStatsCalculation . . . 50
ISongSubmission . . . 35
  SongSubmission . . . 57
IUserLoginDto . . . 36
  AuthController.LoginDto . . . 37
IUserSignupDto . . . 36
IUserStatsCalculation . . . 37
  MongoUserStatsCalculation . . . 51
LoginPage . . . 38

Model 39
Comment 16
Song 54
SongRating 55
User 59
MongoDbContext 40
RateSongComp 52
SearchComponent< TModel > 53
SearchPage 53
SongPage 54
SongRatingItem 56
SongRatingSection 56
SongSearchItem 56
SongStats 57
SongSubmissionPage 58
Thumbnail 58
UserPage 60
UserSearchItem 60
UserSignupForm 61
UserStats 61

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

AuthController
: Controller responsible for authentication operations . . . 15

Comment
: Represents a comment posted by a user on a certain page . . . 16

CommentComp
: A component that displays and manages a single comment, including editing and deletion . . . 17

CommentSection
: A component that displays a list of comments for a given page and allows signed-in users to post a new comment . . . 17

ConfirmDialog
: Dialog component for simple yes/no confirmation . . . 18

DatabaseOperationException
: Represents errors that occur during general database operations . . . 19

Dialog< TResult >
: Base abstract class for all dialogs. Manages opening, closing, and confirmation/cancellation . . . 20

DialogBase
: Provides a HTML wrapper for dialogs with header, body, and footer containing confirm/cancel buttons . . . 21

DialogResult< TResult >
: Represents the result of a dialog . . . 22

Dropzone
: A component that represents a drag-and-drop zone for items. Dropzone must be activated before it can receive items . . . 22

Histogram< TValue, TLabel >
: A generic component that renders a histogram chart based on numeric values and associated labels . . . 23

ICommentsListing
: Provides methods for retrieving comments from the database . . . 24

IDbRepository< TModel >
: Represents a generic repository for performing CRUD operations on models . . . 25

IIdentityProvider
: Provides an abstraction for user authentication, signup, login, logout, and role management . . . 28

InputDialog
: Dialog component that allows user to input a string . . . 30

ISearchOperation< TModel >
: Provides a generic interface for searching entities of type *TModel* . . . 32

ISongRatingListing
Provides methods for retrieving song ratings from the database . . . 33
ISongStatsCalculation
Provides methods for calculating statistics for a song, such as average rating, median rating, total listens, and rating distribution . . . 34
ISongSubmission
Provides an abstraction for submitting new songs to the database while avoiding duplicates . . . 35
IUserLoginDto
Data required to log in a user . . . 36
IUserSignupDto
Data required to sign up a user . . . 36
IUserStatsCalculation
Provides methods for calculating statistics for a user based on their song ratings . . . 37
AuthController.LoginDto
DTO used for login requests . . . 37
LoginPage
Page for users to login or signup . . . 38
Model
Base class for all entities in the system, providing a unique identifier . . . 39
MongoCommentsListing
Provides methods for retrieving comments from the MongoDb database . . . 39
MongoDbContext
Provides a MongoDB client and a method to establish a connection . . . 40
MongoIdentity
A MongoDB-based implementation of IIdentityProvider using ASP.NET Identity. Handles user signup, login, logout, and authentication/authorization checks . . . 41
MongoRepository< TModel >
A MongoDB-based repository for performing CRUD operations on models . . . 44
MongoSongRatingListing
Retrieves song ratings from a MongoDB collection . . . 48
MongoSongSearch
Performs search operations for users in MongoDB . . . 49
MongoSongsStatsCalculation
Calculates statistics for a song, including average rating, median rating, total listens, and rating distribution in MongoDB . . . 50
MongoUserSearch
Performs search operations for songs in MongoDB . . . 51
MongoUserStatsCalculation
Calculates statistics for a user based on their song ratings in MongoDB . . . 51
RateSongComp
A component that allows a signed-in user to rate a song, view their existing rating, update it, or remove it from their collection . . . 52
SearchComponent< TModel >
Base component for search pages. Provides shared search functionality for a given model type 53
SearchPage
A page providing a search interface for songs and users with tabbed navigation . . . 53
Song
A model representing basic song information . . . 54
SongPage
DPage for displaying detailed information about a song, including its metadata, statistics, and comments . . . 54
SongRating
Represents a rating given by a user to a song . . . 55
SongRatingItem
Displays a single song rating item for a user. Can optionally allow inline editing of the rating . . . 56
SongRatingSection
Displays a list of a user's song ratings in descending order. Allows drag-and-drop to reorder ratings if the user is authorized . . . 56

SongSearchItem
Displays an item in search results that navigates to the song's page when clicked 56
SongStats
Represents statistical information about a song, including ratings and listens 57
SongSubmission
Handles submission of new songs to the MongoDB database, ensuring no duplicates exist . . 57
SongSubmissionPage
Page for submitting a new song to the database. Validates user input and handles navigation after submission 58
Thumbnail
A component that displays an image thumbnail or a placeholder if the image link is invalid . . . 58
User
A model representing basic user information 59
UserPage
Displays a user's profile including general info, statistics, and comments. Supports editing the profile for the authorized user 60
UserSearchItem
Displays an item in search results that navigates to the user's page when clicked 60
UserSignupForm
A form component that allows new users to sign up for the application. Handles input validation, submission, and automatic login after successful registration 61
UserStats
Represents statistical information about a user's ratings 61

Chapter 4

Namespace Documentation

4.1 MyMusicTaste Namespace Reference

4.2 MyMusicTaste.Components Namespace Reference

4.3 MyMusicTaste.Components.Comments Namespace Reference

Classes

- class CommentComp

  A component that displays and manages a single comment, including editing and deletion.
- class CommentSection

  A component that displays a list of comments for a given page and allows signed-in users to post a new comment.

4.4 MyMusicTaste.Components.Dialogs Namespace Reference

Classes

- class ConfirmDialog

  Dialog component for simple yes/no confirmation.
- class Dialog< TResult >

  Base abstract class for all dialogs. Manages opening, closing, and confirmation/cancellation.
- class DialogBase

  Provides a HTML wrapper for dialogs with header, body, and footer containing confirm/cancel buttons.
- struct DialogResult< TResult >

  Represents the result of a dialog.
- class InputDialog

  Dialog component that allows user to input a string.

4.5 MyMusicTaste.Components.Misc Namespace Reference

Classes

- class Dropzone

  A component that represents a drag-and-drop zone for items. Dropzone must be activated before it can receive items.
- class Histogram< TValue, TLabel >

  A generic component that renders a histogram chart based on numeric values and associated labels.
- class Thumbnail

  A component that displays an image thumbnail or a placeholder if the image link is invalid.

4.6 MyMusicTaste.Components.Page_Auth Namespace Reference

Classes

- class AuthController

  Controller responsible for authentication operations.
- class LoginPage

  Page for users to login or signup.
- class UserSignupForm

  A form component that allows new users to sign up for the application. Handles input validation, submission, and automatic login after successful registration.

4.7 MyMusicTaste.Components.Page_Home Namespace Reference

4.8 MyMusicTaste.Components.Page_Search Namespace Reference

Classes

- class SearchComponent< TModel >

  Base component for search pages. Provides shared search functionality for a given model type.
- class SearchPage

  A page providing a search interface for songs and users with tabbed navigation.
- class SongSearchItem

  Displays an item in search results that navigates to the song's page when clicked.
- class UserSearchItem

  Displays an item in search results that navigates to the user's page when clicked.

4.9 MyMusicTaste.Components.Page_Song Namespace Reference

Classes

- class RateSongComp

  A component that allows a signed-in user to rate a song, view their existing rating, update it, or remove it from their collection.
- class SongPage

  DPage for displaying detailed information about a song, including its metadata, statistics, and comments.

4.10 MyMusicTaste.Components.Page_SongSubmission Namespace Reference

Classes

- class SongSubmissionPage

  Page for submitting a new song to the database. Validates user input and handles navigation after submission.

4.11 MyMusicTaste.Components.Page_User Namespace Reference

Classes

- class SongRatingItem

  Displays a single song rating item for a user. Can optionally allow inline editing of the rating.
- class SongRatingSection

  Displays a list of a user's song ratings in descending order. Allows drag-and-drop to reorder ratings if the user is authorized.
- class UserPage

  Displays a user's profile including general info, statistics, and comments. Supports editing the profile for the authorized user.

4.12 MyMusicTaste.Database Namespace Reference

Classes

- interface IDbRepository< TModel >

  Represents a generic repository for performing CRUD operations on models.
- interface IIdentityProvider

  Provides an abstraction for user authentication, signup, login, logout, and role management.
- interface IUserLoginDto

  Data required to log in a user.
- interface IUserSignupDto

  Data required to sign up a user.

4.13 MyMusicTaste.Database.Contexts Namespace Reference

4.14 MyMusicTaste.Database.Contexts.MongoDb Namespace Reference

Classes

- class MongoDbContext

  Provides a MongoDB client and a method to establish a connection.
- class MongoIdentity

  A MongoDB-based implementation of IIdentityProvider using ASP.NET Identity. Handles user signup, login, logout, and authentication/authorization checks.
- class MongoRepository< TModel >

  A MongoDB-based repository for performing CRUD operations on models.

4.15 MyMusicTaste.Database.Contexts.MongoDb.Operations Namespace Reference

Classes

- class MongoCommentsListing

  Provides methods for retrieving comments from the MongoDb database.
- class MongoSongRatingListing

  Retrieves song ratings from a MongoDB collection.
- class MongoSongSearch

  Performs search operations for users in MongoDB.
- class MongoSongsStatsCalculation

  Calculates statistics for a song, including average rating, median rating, total listens, and rating distribution in MongoDB.
- class MongoUserSearch

  Performs search operations for songs in MongoDB.
- class MongoUserStatsCalculation

  Calculates statistics for a user based on their song ratings in MongoDB.
- class SongSubmission

  Handles submission of new songs to the MongoDB database, ensuring no duplicates exist.

4.16 MyMusicTaste.Database.Operations Namespace Reference

Classes

- class DatabaseOperationException

  Represents errors that occur during general database operations.
- interface ICommentsListing

  Provides methods for retrieving comments from the database.
- interface ISearchOperation< TModel >

  Provides a generic interface for searching entities of type TModel .
- interface ISongRatingListing

  Provides methods for retrieving song ratings from the database.
- interface ISongStatsCalculation

  Provides methods for calculating statistics for a song, such as average rating, median rating, total listens, and rating distribution.
- interface ISongSubmission

  Provides an abstraction for submitting new songs to the database while avoiding duplicates.
- interface IUserStatsCalculation

  Provides methods for calculating statistics for a user based on their song ratings.

4.17 MyMusicTaste.Models Namespace Reference

Classes

- class Comment

  Represents a comment posted by a user on a certain page.
- class Model

  Base class for all entities in the system, providing a unique identifier.
- class Song

  A model representing basic song information.
- class SongRating

  Represents a rating given by a user to a song.
- class SongStats

  Represents statistical information about a song, including ratings and listens.
- class User

  A model representing basic user information.
- class UserStats

  Represents statistical information about a user's ratings.

Enumerations

- enum CommentPageType { **Song** , **User** }

  Types of pages that a comment can be associated with.

4.18 MyMusicTaste.Startup Namespace Reference

4.19 MyMusicTaste.Utils Namespace Reference

Enumerations

- enum CssUnit {
  Zero , **Px** , **Em** , **Rem** ,
  Vh , **Vw** , **Perc** }

  Represents supported CSS size units.

Functions

- readonly record struct CssSize (double Value, CssUnit Unit)

  Represents a CSS size consisting of a numeric value and a unit. ///.

4.19.1 Function Documentation

4.19.1.1 CssSize()

```
readonly record struct CssSize (
            double Value,
            CssUnit Unit)
```

Parameters

Value	The numeric value of the size.
Unit	The CSS unit of the size.

Converts the size to a CSS value (e.g. "10px", "50%").

Returns

A string representation of the CSS value.

Adds a number to the size value.

Adds a number to the size value.

Adds two sizes with the same unit.

Exceptions

ArgumentException	Thrown when units do not match.

Multiplies the size value by a number.

Multiplies the size value by a number.

Multiplies two sizes with the same unit.

Exceptions

ArgumentException	Thrown when units do not match.

Chapter 5

Class Documentation

5.1 AuthController Class Reference

Controller responsible for authentication operations.

Classes

- class LoginDto

  DTO used for login requests.

Public Member Functions

- **AuthController** (IIdentityProvider identity)
- async Task< IActionResult > LoginAsync ([FromForm] LoginDto loginDto)

  Logs in a user using provided credentials. Returns a redirect to the home page on success, or Unauthorized/Bad↩ Request on failure.
- async Task< IActionResult > LogoutAsync ()

  Logs out the currently authenticated user and redirects to the home page.

Static Public Attributes

- const string **LOGIN_ROUTE** = "auth/login"
- const string **LOGOUT_ROUTE** = "auth/logout"

5.1.1 Member Function Documentation

5.1.1.1 LoginAsync()

```
async Task< IActionResult > LoginAsync (
            [FromForm] LoginDto loginDto)  [inline]
```

Parameters

loginDto	Login credentials.

Returns

An action result representing success or failure of the login attempt.

5.1.1.2 LogoutAsync()

```
async Task< IActionResult > LogoutAsync ()  [inline]
```

Returns

An action result redirecting to the homepage.

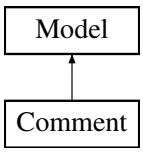
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Auth/AuthController.cs

5.2 Comment Class Reference

Represents a comment posted by a user on a certain page.

Inheritance diagram for Comment:



Properties

- required ObjectId **UserId** `[get, set]`
  The ID of the user who made the comment.
- required CommentPageType **CommentPageType** `[get, set]`
  The type of page the comment was posted in.
- required ObjectId **PageId** `[get, set]`
  The ID of the page the comment was posted to.
- string **Content** = string.Empty `[get, set]`
- DateTime **CreationTime** = INVALID_DATE `[get, set]`
- DateTime **LastEditTime** = INVALID_DATE `[get, set]`
- bool **CreationTimeValid** `[get]`
- bool **LastEditTimeValid** `[get]`

Properties inherited from Model

- ObjectId **Id** `[get, set]`

The documentation for this class was generated from the following file:

- MyMusicTaste/Models/Comment.cs

5.3 CommentComp Class Reference

A component that displays and manages a single comment, including editing and deletion.

Protected Member Functions

- override async Task **OnInitializedAsync** ()

Properties

- Comment **Comment** = null! `[get, set]`

  The comment object to display and manage.
- bool **IsNewComment** = true `[get, set]`

  Indicates whether this comment is newly created and not yet persisted.
- bool **IsPosted** `[get]`

  True if the comment has already been posted to the database.
- EventCallback **OnEditModeOpened** `[get, set]`

  Event triggered when the edit mode is opened.
- EventCallback< bool > **OnEditModeClosed** `[get, set]`

  Event triggered when the edit mode is closed. The callback receives true if changes were saved, false otherwise.
- EventCallback **OnChangesSaved** `[get, set]`

  Event triggered after changes to the comment have been saved.
- EventCallback **OnCommentDeleted** `[get, set]`

  Event triggered after the comment has been deleted.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Comments/CommentComp.razor.cs

5.4 CommentSection Class Reference

A component that displays a list of comments for a given page and allows signed-in users to post a new comment.

Protected Member Functions

- override async Task **OnInitializedAsync** ()

Properties

- CommentPageType **PageType** `[get, set]`

  The type of page the comments was posted to.
- string **PageId** = null! `[get, set]`

  The ID of the page for which to display comments.
- int **ResultsCount** `[get, set]`

  The maximum number of comments to fetch and display.

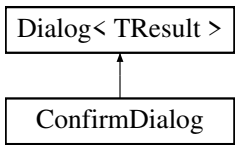
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Comments/CommentSection.razor.cs

5.5 ConfirmDialog Class Reference

Dialog component for simple yes/no confirmation.

Inheritance diagram for ConfirmDialog:



Protected Member Functions

- override bool **GetResult** ()

  Returns the true if the dialog has been confirmed.

Protected Member Functions inherited from Dialog< TResult >

- override void **OnAfterRender** (bool firstRender)
- TResult **GetResult** ()

  Returns the result of the dialog when confirmed.

Properties

- string **Question** = null! `[get, set]`

  Question text to display in the dialog.

Properties inherited from Dialog< TResult >

- string? **Title** `[get, set]`

  Title of the dialog.
- string **ConfirmText** = "Confirm" `[get, set]`

  Text for the confirm button.
- string **CancelText** = "Cancel" `[get, set]`

  Text for the cancel button.
- DialogBase **BaseDialog** = null! `[get, set]`

  Reference to the underlying HTML wrapper dialog component.
- TaskCompletionSource< bool > **CompletionToken** = new() `[get]`

  Token used to track dialog completion.
- Task< bool > **CompletionTask** `[get]`

  Task representing the completion of the dialog.

Additional Inherited Members

Public Member Functions inherited from Dialog< TResult >

- async Task< DialogResult< TResult > > OpenDialog ()

  Opens the dialog and returns a DialogResult<TResult> when confirmed or canceled.
- void **Confirm** ()

  Marks the dialog as confirmed.
- void **Cancel** ()

  Marks the dialog as canceled.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Dialogs/ConfirmDialog.razor.cs

5.6 DatabaseOperationException Class Reference

Represents errors that occur during general database operations.

Public Member Functions

- **DatabaseOperationException** (string message)
- **DatabaseOperationException** (string message, Exception innerException)

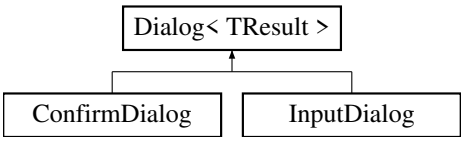
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Operations/DatabaseOperationException.cs

5.7 Dialog< TResult > Class Template Reference

Base abstract class for all dialogs. Manages opening, closing, and confirmation/cancellation.

Inheritance diagram for Dialog< TResult >:



Public Member Functions

- async Task< DialogResult< TResult > > OpenDialog ()

  Opens the dialog and returns a DialogResult< TResult > when confirmed or canceled.
- void **Confirm** ()

  Marks the dialog as confirmed.
- void **Cancel** ()

  Marks the dialog as canceled.

Protected Member Functions

- override void **OnAfterRender** (bool firstRender)
- TResult **GetResult** ()

  Returns the result of the dialog when confirmed.

Properties

- string? **Title** `[get, set]`

  Title of the dialog.
- string **ConfirmText** = "Confirm" `[get, set]`

  Text for the confirm button.
- string **CancelText** = "Cancel" `[get, set]`

  Text for the cancel button.
- DialogBase **BaseDialog** = null! `[get, set]`

  Reference to the underlying HTML wrapper dialog component.
- TaskCompletionSource< bool > **CompletionToken** = new() `[get]`

  Token used to track dialog completion.
- Task< bool > **CompletionTask** `[get]`

  Task representing the completion of the dialog.

5.7.1 Detailed Description

Template Parameters

TResult	The type of the result returned when the dialog is confirmed.

5.7.2 Member Function Documentation

5.7.2.1 OpenDialog()

```
async Task< DialogResult< TResult > > OpenDialog ()  [inline]
```

Returns

A task for the result of the dialog

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Dialogs/Dialog.cs

5.8 DialogBase Class Reference

Provides a HTML wrapper for dialogs with header, body, and footer containing confirm/cancel buttons.

Public Member Functions

- void **Open** ()
  Makes the dialog visible.
- void **Close** ()
  Makes the dialog invisible.

Properties

- RenderFragment **BodyContent** = null! `[get, set]`
  Content to display inside the dialog body.
- string? **Title** `[get, set]`
  Title of the dialog.
- string **ConfirmText** = "Yes" `[get, set]`
  Text for the confirm button.
- string **CancelText** = "No" `[get, set]`
  Text for the cancel button.
- bool **IsVisible** `[get]`
  Indicates whether the dialog is currently visible.

Events

- Action? **ConfirmPressed**

  Event triggered when the confirm button is pressed.
- Action? **CancelPressed**

  Event triggered when the cancel button is pressed.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Dialogs/DialogBase.razor.cs

5.9 DialogResult< TResult > Struct Template Reference

Represents the result of a dialog.

Properties

- bool **WasConfirmed** `[get, set]`

  Indicates if the dialog was confirmed.
- TResult? **Result** `[get, set]`

  The value returned if the dialog was confirmed; otherwise default.

5.9.1 Detailed Description

Template Parameters

TResult	Type of the value returned when confirmed.

The documentation for this struct was generated from the following file:

- MyMusicTaste/Components/Dialogs/DialogResult.cs

5.10 Dropzone Class Reference

A component that represents a drag-and-drop zone for items. Dropzone must be activated before it can receive items.

Public Member Functions

- void SetActive (bool active)

  Sets the active state of the dropzone.

Properties

- bool **AutoActivate** = true [get, set]

  If true, the dropzone automatically activates when an item is dragged over it.
- bool **AutoDeactivate** = true [get, set]

  If true, the dropzone automatically deactivates when an item leaves it.
- EventCallback< Dropzone > **OnDragEnter** [get, set]

  Event triggered when an item is dragged into the dropzone.
- EventCallback< Dropzone > **OnDragLeave** [get, set]

  Event triggered when an item leaves the dropzone.
- EventCallback< Dropzone > **OnDrop** [get, set]

  Event triggered when an item is dropped onto the dropzone.

5.10.1 Member Function Documentation

5.10.1.1 SetActive()

```
void SetActive (
            bool active)  [inline]
```

Parameters

active	True to activate, false to deactivate.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Misc/Dropzone.razor.cs

5.11 Histogram< TValue, TLabel > Class Template Reference

A generic component that renders a histogram chart based on numeric values and associated labels.

Protected Member Functions

- override void **OnInitialized** ()

Properties

- TValue[ ] **Values** = null! [get, set]

  Array of numeric values to display in the histogram.
- TLabel[ ] **Labels** = null! [get, set]

  Array of labels corresponding to the values.
- CssSize **MaxHeight** [get, set]

  Maximum height of the histogram bars.

5.11.1 Detailed Description

Template Parameters

TValue	The numeric type of the values to display.
TLabel	The type of labels for each value.

Type Constraints

TValue **:** ***INumber*<*TValue*>**

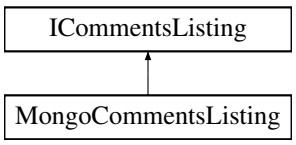
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Misc/Histogram.razor.cs

5.12 ICommentsListing Interface Reference

Provides methods for retrieving comments from the database.

Inheritance diagram for ICommentsListing:



Public Member Functions

- Task< IEnumerable< Comment > > GetCommentsByUserAsync (string userId, int resultCount)

  Retrieves comments made by a specific user.
- Task< IEnumerable< Comment > > GetCommentsByPageAsync (CommentPageType pageType, string pageId, int resultCount)

  Retrieves comments associated with a specific page, ordered by creation time descending.

5.12.1 Member Function Documentation

5.12.1.1 GetCommentsByPageAsync()

```
Task< IEnumerable< Comment > > GetCommentsByPageAsync (
            CommentPageType pageType,
            string pageId,
            int resultCount)
```

Parameters

pageType	The type of the page the comments belong to.
pageId	The ID of the page.
resultCount	The maximum number of comments to return.

Returns

A task for the list of comments.

Implemented in MongoCommentsListing.

5.12.1.2 GetCommentsByUserAsync()

```
Task< IEnumerable< Comment > > GetCommentsByUserAsync (
            string userId,
            int resultCount)
```

Parameters

userId	The ID of the user whose comments to retrieve.
resultCount	The maximum number of comments to return.

Returns

A task for the collection of comments.

Implemented in MongoCommentsListing.

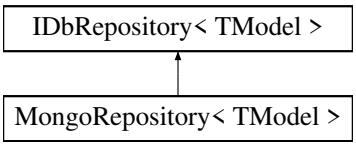
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/Operations/ICommentsListing.cs

5.13 IDbRepository< TModel > Interface Template Reference

Represents a generic repository for performing CRUD operations on models.

Inheritance diagram for IDbRepository< TModel >:



Public Member Functions

- TModel GetById (string? id)

  Retrieves a model by its ID from the database.
- TModel GetById (ObjectId id)

  Retrieves a model by its ID from the database.
- Task< TModel > GetByIdAsync (string? id)

  Retrieves a model by its string ID from the database.
- Task< IEnumerable< TModel > > GetByIdsAsync (IEnumerable< string > ids)

  Asynchronously retrieves multiple models by their IDs form the database.
- Task CreateAsync (TModel model)

  Asynchronously inserts a new model to the database.
- Task UpdateAsync (TModel model)

  Asynchronously updates an existing model in the database.
- Task DeleteAsync (TModel model)

  Asynchronously deletes a model from the database.

5.13.1 Detailed Description

Template Parameters

TModel	The type of model managed by this repository. Must inherit from Model.

5.13.2 Member Function Documentation

5.13.2.1 CreateAsync()

```
Task CreateAsync (
            TModel model)
```

Parameters

model	The model to insert.

Implemented in MongoRepository< TModel >.

5.13.2.2 DeleteAsync()

```
Task DeleteAsync (
            TModel model)
```

Parameters

model	The model to delete.

Implemented in MongoRepository< TModel >.

5.13.2.3 GetById() [1/2]

```
TModel GetById (
            ObjectId id)
```

Parameters

id	The unique identifier of the model.

Returns

The model corresponding to the specified ID.

Exceptions

EntryNotFoundException	Thrown when the entry does not exist.

Implemented in MongoRepository< TModel >.

5.13.2.4 GetById() [2/2]

```
TModel GetById (
            string?  id)
```

Parameters

id	The unique identifier of the model.

Returns

The model corresponding to the specified ID.

Exceptions

EntryNotFoundException	Thrown when the entry does not exist.

Implemented in MongoRepository< TModel >.

5.13.2.5 GetByIdAsync()

```
Task< TModel > GetByIdAsync (
            string?  id)
```

Parameters

id	The unique identifier of the model.

Returns

A task for the retrieved model.

Exceptions

EntryNotFoundException	Thrown when the entry does not exist.

Implemented in MongoRepository< TModel >.

5.13.2.6 GetByIdsAsync()

```
Task< IEnumerable< TModel > > GetByIdsAsync (
            IEnumerable< string > ids)
```

Parameters

ids	The collection of unique identifiers to retrieve.

Returns

A task for the collection of retrieved models.

Implemented in MongoRepository< TModel >.

5.13.2.7 UpdateAsync()

```
Task UpdateAsync (
            TModel model)
```

Parameters

model	The model with updated values.

Implemented in MongoRepository< TModel >.

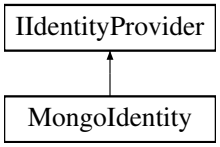
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/IDbRepository.cs

5.14 IIdentityProvider Interface Reference

Provides an abstraction for user authentication, signup, login, logout, and role management.

Inheritance diagram for IIdentityProvider:



Public Member Functions

- Task< IdentityResult > SignUpUserAsync (IUserSignupDto userSignup)
  Signs up a new user.
- Task< SignInResult > LoginUserAsync (IUserLoginDto userLoginDto)
  Logs in a user.
- Task LogOutUserAsync ()
  Logs out the currently signed-in user.
- Task< IdentityResult > AssignRoleAsync (IdentityUser user, IdentityRole role)
  Assigns a role to a user.
- bool IsAuthenticated ()
  Checks if a user is authenticated.
- bool AuthorizeUserById (string requiredUserId)
  Checks if a signed-in user has a specific ID.
- string? GetUserId ()
  Gets the currently signed-in user's ID.

5.14.1 Member Function Documentation

5.14.1.1 AssignRoleAsync()

```
Task< IdentityResult > AssignRoleAsync (
            IdentityUser user,
            IdentityRole role)
```

Parameters

user	The user to assign the role to.
role	The role to assign.

Returns

A task for the IdentityResult.

Implemented in MongoIdentity.

5.14.1.2 AuthorizeUserById()

```
bool AuthorizeUserById (
            string requiredUserId)
```

Parameters

requiredUserId	The required user ID.

Returns

True if the user has the specified ID; otherwise false.

Implemented in MongoIdentity.

5.14.1.3 GetUserId()

```
string?  GetUserId ()
```

Returns

The user ID as a string, or null if not authenticated.

Implemented in MongoIdentity.

5.14.1.4 IsAuthenticated()

```
bool IsAuthenticated ()
```

Returns

True if a user is authenticated; otherwise false.

Implemented in MongoIdentity.

5.14.1.5 LoginUserAsync()

```
Task< SignInResult > LoginUserAsync (
            IUserLoginDto userLoginDto)
```

Parameters

userLoginDto	The user login data.

Returns

A task for the SignInResult indicating success or failure.

Implemented in MongoIdentity.

5.14.1.6 LogOutUserAsync()

```
Task LogOutUserAsync ()
```

Implemented in MongoIdentity.

5.14.1.7 SignUpUserAsync()

```
Task< IdentityResult > SignUpUserAsync (
            IUserSignupDto userSignup)
```

Parameters

userSignup	The user signup data.

Returns

A task for the IdentityResult indicating success or failure.

Implemented in MongoIdentity.

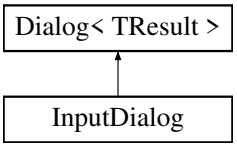
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/IIdentityProvider.cs

5.15 InputDialog Class Reference

Dialog component that allows user to input a string.

Inheritance diagram for InputDialog:



Protected Member Functions

- override string **GetResult** ()

  Returns the text entered in the dialog.

Protected Member Functions inherited from Dialog< TResult >

- override void **OnAfterRender** (bool firstRender)
- TResult **GetResult** ()

  Returns the result of the dialog when confirmed.

Properties

- string **Placeholder** = "" `[get, set]`

  Placeholder text displayed inside the input area.

Properties inherited from Dialog< TResult >

- string? **Title** `[get, set]`

  Title of the dialog.
- string **ConfirmText** = "Confirm" `[get, set]`

  Text for the confirm button.
- string **CancelText** = "Cancel" `[get, set]`

  Text for the cancel button.
- DialogBase **BaseDialog** = null! `[get, set]`

  Reference to the underlying HTML wrapper dialog component.
- TaskCompletionSource< bool > **CompletionToken** = new() `[get]`

  Token used to track dialog completion.
- Task< bool > **CompletionTask** `[get]`

  Task representing the completion of the dialog.

Additional Inherited Members

Public Member Functions inherited from Dialog< TResult >

- async Task< DialogResult< TResult > > OpenDialog ()

  Opens the dialog and returns a DialogResult<TResult> when confirmed or canceled.
- void **Confirm** ()

  Marks the dialog as confirmed.
- void **Cancel** ()

  Marks the dialog as canceled.

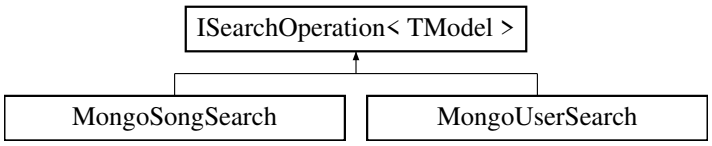
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Dialogs/InputDialog.razor.cs

5.16 ISearchOperation< TModel > Interface Template Reference

Provides a generic interface for searching entities of type *TModel* .

Inheritance diagram for ISearchOperation< TModel >:



Public Member Functions

- Task< IEnumerable< TModel >?> SearchAsync (string query, int resultsCount)
  Searches entities based on a query string.

5.16.1 Detailed Description

Template Parameters

TModel	The type of model to search for.

Type Constraints

TModel* : *Model

5.16.2 Member Function Documentation

5.16.2.1 SearchAsync()

```
Task< IEnumerable< TModel >?> SearchAsync (
            string query,
            int resultsCount)
```

Parameters

query	The search query string.
resultsCount	The maximum number of results to return.

Returns

A task for the collection of matching entities, or null if the query is empty.

Implemented in MongoSongSearch, and MongoUserSearch.

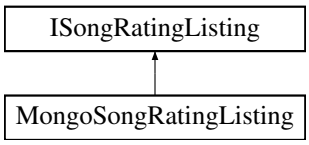
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/Operations/ISearchOperation.cs

5.17 ISongRatingListing Interface Reference

Provides methods for retrieving song ratings from the database.

Inheritance diagram for ISongRatingListing:



Public Member Functions

- Task< SongRating > GetSongRatingAsync (string songId, string userId)

  Retrieves the rating a specific user has given to a specific song.
- Task< IEnumerable< SongRating > > GetRatingsByUserAsync (string userId)

  Retrieves all ratings submitted by a specific user.
- Task< IEnumerable< SongRating > > GetRatingsBySongAsync (string songId)

  Retrieves all ratings associated with a specific song.

5.17.1 Member Function Documentation

5.17.1.1 GetRatingsBySongAsync()

```
Task< IEnumerable< SongRating > > GetRatingsBySongAsync (
            string songId)
```

Parameters

songId	The ID of the song.

Returns

A task for the list of ratings.

Implemented in MongoSongRatingListing.

5.17.1.2 GetRatingsByUserAsync()

```
Task< IEnumerable< SongRating > > GetRatingsByUserAsync (
            string userId)
```

Parameters

user↩ Id	The ID of the user.

Returns

A task for the list of ratings.

Implemented in MongoSongRatingListing.

5.17.1.3 GetSongRatingAsync()

```
Task< SongRating > GetSongRatingAsync (
            string songId,
            string userId)
```

Parameters

song↩ Id	The ID of the song.
userId	The ID of the user.

Returns

A task for the retrieved song rating, or null if no rating exists.

Implemented in MongoSongRatingListing.

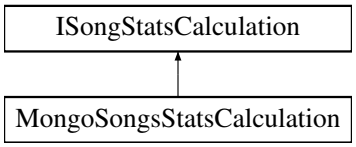
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/Operations/ISongRatingListing.cs

5.18 ISongStatsCalculation Interface Reference

Provides methods for calculating statistics for a song, such as average rating, median rating, total listens, and rating distribution.

Inheritance diagram for ISongStatsCalculation:



Public Member Functions

- Task< SongStats > CalculateSongStatsAsync (Song song)
  Calculates statistics for a song.

5.18.1 Member Function Documentation

5.18.1.1 CalculateSongStatsAsync()

```
Task< SongStats > CalculateSongStatsAsync (
            Song song)
```

Parameters

song	The song to calculate statistics for.

Returns

A task for the song's statistics.

Implemented in MongoSongsStatsCalculation.

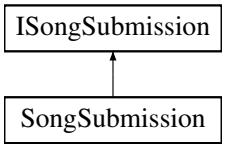
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/Operations/ISongStatsCalculation.cs

5.19 ISongSubmission Interface Reference

Provides an abstraction for submitting new songs to the database while avoiding duplicates.

Inheritance diagram for ISongSubmission:



Public Member Functions

- Task< string > SubmitSongAsync (Song songModel)
  Submits a new song.

5.19.1 Member Function Documentation

5.19.1.1 SubmitSongAsync()

```
Task< string > SubmitSongAsync (
            Song songModel)
```

Parameters

songModel	The song to submit.

Returns

A task for the ID of the newly created song as a string.

Exceptions

EntryAlreadyExistsException	Thrown if the song already exists.

Implemented in SongSubmission.

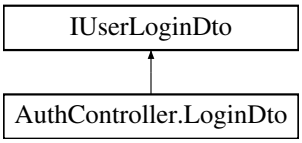
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/Operations/ISongSubmission.cs

5.20 IUserLoginDto Interface Reference

Data required to log in a user.

Inheritance diagram for IUserLoginDto:



Properties

- string **Username** `[get, set]`
- string **Password** `[get, set]`

The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/IIdentityProvider.cs

5.21 IUserSignupDto Interface Reference

Data required to sign up a user.

Properties

- string **Username** `[get, set]`
- string **Email** `[get, set]`
- string **Password** `[get, set]`

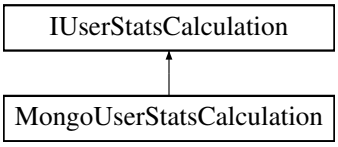
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/IIdentityProvider.cs

5.22 IUserStatsCalculation Interface Reference

Provides methods for calculating statistics for a user based on their song ratings.

Inheritance diagram for IUserStatsCalculation:



Public Member Functions

- Task< UserStats > CalculateUserStatsAsync (string userId)

  Calculates statistics for a user, such as average ratings grouped by album, author, and genre.

5.22.1 Member Function Documentation

5.22.1.1 CalculateUserStatsAsync()

```
Task< UserStats > CalculateUserStatsAsync (
            string userId)
```

Parameters

userId	The ID of the user to calculate statistics for.

Returns

A task for the user's statistics.

Implemented in MongoUserStatsCalculation.

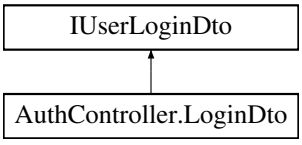
The documentation for this interface was generated from the following file:

- MyMusicTaste/Database/Operations/IUserStatsCalculation.cs

5.23 AuthController.LoginDto Class Reference

DTO used for login requests.

Inheritance diagram for AuthController.LoginDto:



Properties

- string Username = null! `[get, set]`
- string Password = null! `[get, set]`

5.23.1 Property Documentation

5.23.1.1 Password

```
string Password = null!  [get], [set]
```

Implements IUserLoginDto.

5.23.1.2 Username

```
string Username = null!  [get], [set]
```

Implements IUserLoginDto.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Auth/AuthController.cs

5.24 LoginPage Class Reference

Page for users to login or signup.

Static Public Member Functions

- static string **GetRoute** ()

Static Public Attributes

- const string **ROUTE_TEMPLATE** = "/login"

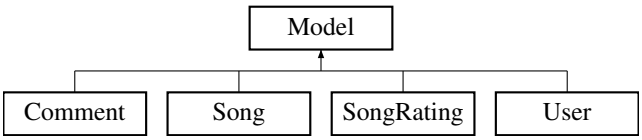
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Auth/LoginPage.razor.cs

5.25 Model Class Reference

Base class for all entities in the system, providing a unique identifier.

Inheritance diagram for Model:



Properties

- ObjectId **Id** `[get, set]`

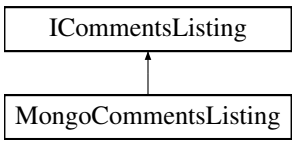
The documentation for this class was generated from the following file:

- MyMusicTaste/Models/Model.cs

5.26 MongoCommentsListing Class Reference

Provides methods for retrieving comments from the MongoDb database.

Inheritance diagram for MongoCommentsListing:



Public Member Functions

- Task< IEnumerable< Comment > > GetCommentsByUserAsync (string userId, int resultCount)

  Retrieves comments made by a specific user.
- async Task< IEnumerable< Comment > > GetCommentsByPageAsync (CommentPageType pageType, string pageId, int resultCount)

  Retrieves comments associated with a specific page, ordered by creation time descending.

5.26.1 Member Function Documentation

5.26.1.1 GetCommentsByPageAsync()

```
async Task< IEnumerable< Comment > > GetCommentsByPageAsync (
            CommentPageType pageType,
            string pageId,
            int resultCount)  [inline]
```

Parameters

pageType	The type of the page the comments belong to.
pageId	The ID of the page.
resultCount	The maximum number of comments to return.

Returns

A task for the collection of comments.

Implements ICommentsListing.

5.26.1.2 GetCommentsByUserAsync()

```
Task< IEnumerable< Comment > > GetCommentsByUserAsync (
            string userId,
            int resultCount)  [inline]
```

Parameters

userId	The ID of the user whose comments to retrieve.
resultCount	The maximum number of comments to return.

Returns

A task for the list of comments.

Implements ICommentsListing.

The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoCommentsListing.cs

5.27 MongoDbContext Class Reference

Provides a MongoDB client and a method to establish a connection.

Static Public Member Functions

- static void Connect (string? key)

  Connects to the MongoDB database using the provided connection string.

Properties

- static ? MongoClient **Client** `[get]`

  The singleton MongoDB client instance.

5.27.1 Member Function Documentation

5.27.1.1 Connect()

```
void Connect (
            string?  key)  [inline], [static]
```

Parameters

key	The MongoDB connection string.

Exceptions

ArgumentException	Thrown if the connection string is null or empty.

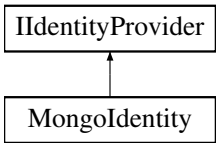
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/MongoDbContext.cs

5.28 MongoIdentity Class Reference

A MongoDB-based implementation of IIdentityProvider using ASP.NET Identity. Handles user signup, login, logout, and authentication/authorization checks.

Inheritance diagram for MongoIdentity:



Public Member Functions

- MongoIdentity (SignInManager< MongoAccount > signInManager, IDbRepository< User > userRepository, IHttpContextAccessor httpContextAccessor)

  Initializes a new instance of MongoIdentity with required ASP.NET Identity services.
- async Task< IdentityResult > SignUpUserAsync (IUserSignupDto newUserSignup)

  Signs up a new user and stores their account in MongoDB.
- async Task< SignInResult > LoginUserAsync (IUserLoginDto userLogin)

  Logs in a user using ASP.NET Identity sign-in manager.
- async Task LogOutUserAsync ()

  Logs out the currently signed-in user.
- Task< IdentityResult > AssignRoleAsync (IdentityUser user, IdentityRole role)

  Assigns a role to a user. (Not implemented in this MongoDB version.)
- bool IsAuthenticated ()

  Checks if the current HTTP context has an authenticated user.
- bool AuthorizeUserById (string requiredUserId)

  Checks if the current user has a specific user ID claim.
- string? GetUserId ()

  Retrieves the current user's ID from the HTTP context.

Static Public Member Functions

- static void Configure (IServiceCollection services, string connectionString)

  Configures ASP.NET Identity services and registers MongoIdentity as the identity provider.

5.28.1 Constructor & Destructor Documentation

5.28.1.1 MongoIdentity()

```
MongoIdentity (
            SignInManager< MongoAccount > signInManager,
            IDbRepository< User > userRepository,
            IHttpContextAccessor httpContextAccessor)  [inline]
```

Parameters

signInManager	The SignInManager for user login/logout.
userRepository	Repository for storing application user data.
httpContextAccessor	Accessor for the current HTTP context.

5.28.2 Member Function Documentation

5.28.2.1 AssignRoleAsync()

```
Task< IdentityResult > AssignRoleAsync (
            IdentityUser user,
            IdentityRole role)  [inline]
```

Parameters

user	The identity user to assign the role to.
role	The identity role to assign.

Returns

A task for the IdentityResult.

Implements IIdentityProvider.

5.28.2.2 AuthorizeUserById()

```
bool AuthorizeUserById (
            string requiredUserId)  [inline]
```

Parameters

requiredUserId	The required user ID.

Returns

True if the user has the claim; otherwise false.

Implements IIdentityProvider.

5.28.2.3 Configure()

```
void Configure (
            IServiceCollection services,
            string connectionString)  [inline], [static]
```

Parameters

services	The service collection to configure.
connectionString	The MongoDB connection string for the identity store.

5.28.2.4 GetUserId()

```
string?  GetUserId ()  [inline]
```

Returns

The user ID as a string, or null if not authenticated.

Implements IIdentityProvider.

5.28.2.5 IsAuthenticated()

```
bool IsAuthenticated ()  [inline]
```

Returns

True if a user is authenticated; otherwise false.

Implements IIdentityProvider.

5.28.2.6 LoginUserAsync()

```
async Task< SignInResult > LoginUserAsync (
            IUserLoginDto userLogin)  [inline]
```

Parameters

userLogin	The user login data transfer object.

Returns

A task for the SignInResult indicating success or failure.

Implements IIdentityProvider.

5.28.2.7 LogOutUserAsync()

```
async Task LogOutUserAsync ()  [inline]
```

Implements IIdentityProvider.

5.28.2.8 SignUpUserAsync()

```
async Task< IdentityResult > SignUpUserAsync (
            IUserSignupDto newUserSignup)  [inline]
```

Parameters

newUserSignup	The user signup data transfer object.

Returns

A task for the IdentityResult indicating success or failure.

Implements IIdentityProvider.

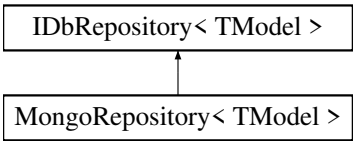
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/MongoIdentity.cs

5.29 MongoRepository< TModel > Class Template Reference

A MongoDB-based repository for performing CRUD operations on models.

Inheritance diagram for MongoRepository< TModel >:



Public Member Functions

- TModel GetById (string? id)

  Retrieves a model by its ID.
- TModel GetById (ObjectId id)

  Retrieves a model by its MongoDB ObjectId.
- async Task< TModel > GetByIdAsync (string? id)

  Asynchronously retrieves a model by its ID.
- async Task< IEnumerable< TModel > > GetByIdsAsync (IEnumerable< string > ids)

  Asynchronously retrieves multiple models by their IDs.
- async Task CreateAsync (TModel model)

  Asynchronously inserts a new model into the database collection.
- async Task UpdateAsync (TModel model)

  Asynchronously updates an existing model in the database collection.
- async Task DeleteAsync (TModel model)

  Asynchronously deletes a model from the collection.

Properties

- IMongoCollection< TModel > **Collection** = MongoCollectionFactory.Create<TModel>() `[get]`

  The MongoDB collection associated with the model type.

5.29.1 Detailed Description

Template Parameters

TModel	The type of model managed by this repository. Must inherit from Model.

Type Constraints

TModel* : *Model

5.29.2 Member Function Documentation

5.29.2.1 CreateAsync()

```
async Task CreateAsync (
            TModel model)  [inline]
```

Parameters

model	The model to insert.

Implements IDbRepository< TModel >.

5.29.2.2 DeleteAsync()

```
async Task DeleteAsync (
            TModel model)  [inline]
```

Parameters

model	The model to delete.

Implements IDbRepository< TModel >.

5.29.2.3 GetById() [1/2]

```
TModel GetById (
            ObjectId id)  [inline]
```

Parameters

id	The MongoDB ObjectId of the model.

Returns

The model corresponding to the specified ID.

Exceptions

EntryNotFoundException	Thrown when no entry is found.

Implements IDbRepository< TModel >.

5.29.2.4 GetById() [2/2]

```
TModel GetById (
            string?  id)  [inline]
```

Parameters

id	The string representation of the model's ObjectId.

Returns

The model corresponding to the specified ID.

Exceptions

EntryNotFoundException	Thrown when the ID is invalid or no entry is found.

Implements IDbRepository< TModel >.

5.29.2.5 GetByIdAsync()

```
async Task< TModel > GetByIdAsync (
            string?  id)  [inline]
```

Parameters

id	The string representation of the model's ObjectId.

Returns

A task for the model corresponding to the specified ID.

Exceptions

EntryNotFoundException	Thrown when no entry is found.

Implements IDbRepository< TModel >.

5.29.2.6 GetByIdsAsync()

```
async Task< IEnumerable< TModel > > GetByIdsAsync (
            IEnumerable< string > ids)  [inline]
```

Parameters

ids	The collection of string IDs to retrieve.

Returns

A task for the collection of matching models.

Implements IDbRepository< TModel >.

5.29.2.7 UpdateAsync()

```
async Task UpdateAsync (
            TModel model)  [inline]
```

Parameters

model	The model with updated values.

Implements IDbRepository< TModel >.

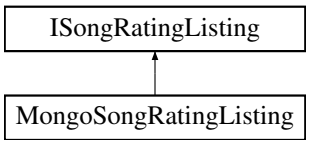
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/MongoRepository.cs

5.30 MongoSongRatingListing Class Reference

Retrieves song ratings from a MongoDB collection.

Inheritance diagram for MongoSongRatingListing:



Public Member Functions

- Task< SongRating > GetSongRatingAsync (string songId, string userId)
  Retrieves the rating a specific user has given to a specific song.
- async Task< IEnumerable< SongRating > > GetRatingsByUserAsync (string userId)
  Retrieves all ratings submitted by a specific user.
- async Task< IEnumerable< SongRating > > GetRatingsBySongAsync (string songId)
  Retrieves all ratings associated with a specific song.

5.30.1 Member Function Documentation

5.30.1.1 GetRatingsBySongAsync()

```
async Task< IEnumerable< SongRating > > GetRatingsBySongAsync (
            string songId)  [inline]
```

Parameters

songId	The ID of the song.

Returns

A task for the collection of ratings.

Implements ISongRatingListing.

5.30.1.2 GetRatingsByUserAsync()

```
async Task< IEnumerable< SongRating > > GetRatingsByUserAsync (
            string userId)  [inline]
```

Parameters

user↩ Id	The ID of the user.

Returns

A task for the collection of ratings.

Implements ISongRatingListing.

5.30.1.3 GetSongRatingAsync()

```
Task< SongRating > GetSongRatingAsync (
            string songId,
            string userId)  [inline]
```

Parameters

song↩ Id	The ID of the song.
userId	The ID of the user.

Returns

A task for the retrieved song rating, or null if no rating exists.

Implements ISongRatingListing.

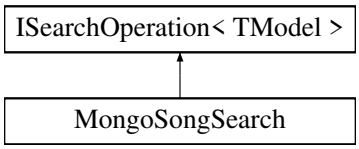
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoSongRatingListing.cs

5.31 MongoSongSearch Class Reference

Performs search operations for users in MongoDB.

Inheritance diagram for MongoSongSearch:



Public Member Functions

- async Task< IEnumerable< Song >?> SearchAsync (string query, int resultsCount)
  Searches users by username using autocomplete.

5.31.1 Member Function Documentation

5.31.1.1 SearchAsync()

```
async Task< IEnumerable< Song >?> SearchAsync (
            string query,
            int resultsCount)  [inline]
```

Parameters

query	The search query string.
resultsCount	The maximum number of results to return.

Returns

A task for the collection of matching users, or null if the query is empty.

Implements ISearchOperation< TModel >.

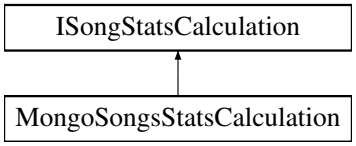
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoSongSearch.cs

5.32 MongoSongsStatsCalculation Class Reference

Calculates statistics for a song, including average rating, median rating, total listens, and rating distribution in MongoDB.

Inheritance diagram for MongoSongsStatsCalculation:



Public Member Functions

- async Task< SongStats > CalculateSongStatsAsync (Song song)

  Calculates statistics for a song, including average rating, median rating, total listens, and rating distribution.

5.32.1 Member Function Documentation

5.32.1.1 CalculateSongStatsAsync()

```
async Task< SongStats > CalculateSongStatsAsync (
            Song song)  [inline]
```

Parameters

song	The song to calculate statistics for.

Returns

A task for the song's statistics.

Implements ISongStatsCalculation.

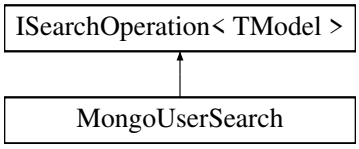
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoSongsStatsCalculation.cs

5.33 MongoUserSearch Class Reference

Performs search operations for songs in MongoDB.

Inheritance diagram for MongoUserSearch:



Public Member Functions

- async Task< IEnumerable< User >?> SearchAsync (string query, int resultsCount)
  Searches songs by title using autocomplete.

5.33.1 Member Function Documentation

5.33.1.1 SearchAsync()

```
async Task< IEnumerable< User >?> SearchAsync (
            string query,
            int resultsCount)  [inline]
```

Parameters

query	The search query string.
resultsCount	The maximum number of results to return.

Returns

A task for the collection of matching songs, or null if the query is empty.

Implements ISearchOperation< TModel >.

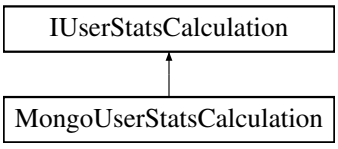
The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoUserSearch.cs

5.34 MongoUserStatsCalculation Class Reference

Calculates statistics for a user based on their song ratings in MongoDB.

Inheritance diagram for MongoUserStatsCalculation:



Public Member Functions

- async Task< UserStats > CalculateUserStatsAsync (string userId)

  Calculates statistics for a user based on their ratings, including average ratings by album, author, and genre.

5.34.1 Member Function Documentation

5.34.1.1 CalculateUserStatsAsync()

```
async Task< UserStats > CalculateUserStatsAsync (
            string userId)  [inline]
```

Parameters

user↩ Id	The ID of the user to calculate statistics for.

Returns

A task for the user's statistics.

Implements IUserStatsCalculation.

The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoUserStatsCalculation.cs

5.35 RateSongComp Class Reference

A component that allows a signed-in user to rate a song, view their existing rating, update it, or remove it from their collection.

Protected Member Functions

- override async Task **OnInitializedAsync** ()

Properties

- string **SongId** = null! `[get, set]`

  The ID of the song to rate.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Song/RateSongComp.razor.cs

5.36 SearchComponent< TModel > Class Template Reference

Base component for search pages. Provides shared search functionality for a given model type.

Public Member Functions

- async Task **UpdateResults** ()

  Performs the search with the current query and updates the Results.

Protected Member Functions

- override async Task **OnAfterRenderAsync** (bool _)

Properties

- string? **Query** `[get, set]`

  The search query string entered by the user.
- int **ResultsCount** `[get, set]`

  Maximum number of results to return.
- ISearchOperation< TModel > **Searcher** = null! `[get, set]`

  The search operation service used to perform the query.
- IEnumerable< TModel >? **Results** `[get, set]`

  The list of search results returned by the query.

5.36.1 Detailed Description

Template Parameters

TModel	The type of the model to search for, must inherit from Model.

Type Constraints

TModel* : *Model

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Search/SearchComponent.cs

5.37 SearchPage Class Reference

A page providing a search interface for songs and users with tabbed navigation.

Static Public Member Functions

- static string **GetRoute** ()

Static Public Attributes

- const string **ROUTE_TEMPLATE** = "/search"

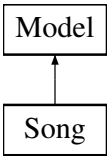
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Search/SearchPage.razor.cs

5.38 Song Class Reference

A model representing basic song information.

Inheritance diagram for Song:



Properties

- string **Title** = null! `[get, set]`
- string **Author** = null! `[get, set]`
- string? **Album** `[get, set]`
- string? **Genre** `[get, set]`
- DateOnly **ReleaseDate** = INVALID_DATE `[get, set]`
- bool **ReleasDateValid** `[get]`
- string? **CoverImageLink** `[get, set]`
- string? **SourceLink** `[get, set]`

Properties inherited from Model

- ObjectId **Id** `[get, set]`

The documentation for this class was generated from the following file:

- MyMusicTaste/Models/Song/Song.cs

5.39 SongPage Class Reference

DPage for displaying detailed information about a song, including its metadata, statistics, and comments.

Static Public Member Functions

- static string **GetRoute** (string songId)

Static Public Attributes

- const string **ROUTE_TEMPLATE** = "/songs/{SongId}"

Protected Member Functions

- override async Task **OnInitializedAsync** ()

Properties

- string **SongId** = null! `[get, set]`
  The ID of the song to display.

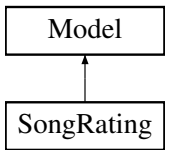
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Song/SongPage.razor.cs

5.40 SongRating Class Reference

Represents a rating given by a user to a song.

Inheritance diagram for SongRating:



Static Public Attributes

- const byte **NOT_RATED** = 255
  Sentinel value of the Rating field, indicating that a song has not been rated.

Properties

- ObjectId **UserId** `[get, set]`
- ObjectId **SongId** `[get, set]`
- byte **Rating** = NOT_RATED `[get, set]`
- bool **IsRated** `[get]`

Properties inherited from Model

- ObjectId **Id** `[get, set]`

The documentation for this class was generated from the following file:

- MyMusicTaste/Models/User/SongRating.cs

5.41 SongRatingItem Class Reference

Displays a single song rating item for a user. Can optionally allow inline editing of the rating.

Protected Member Functions

- override void **OnInitialized** ()

Properties

- SongRating **Rating** = new() `[get, set]`
  The rating object of the item.
- bool **Editable** `[get, set]`
  Determines whether the rating can be edited inline.
- EventCallback **OnRatingSaved** `[get, set]`
  Invoked after a rating has been saved.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-User/SongRatingItem.razor.cs

5.42 SongRatingSection Class Reference

Displays a list of a user's song ratings in descending order. Allows drag-and-drop to reorder ratings if the user is authorized.

Protected Member Functions

- override async Task **OnInitializedAsync** ()

Properties

- string **UserId** = null! `[get, set]`
  The ID of the user whose ratings will be displayed.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-User/SongRatingSection.razor.cs

5.43 SongSearchItem Class Reference

Displays an item in search results that navigates to the song's page when clicked.

Properties

- Models.? Song **Song** = new() `[get, set]`

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Search/SongSearchItem.razor.cs

5.44 SongStats Class Reference

Represents statistical information about a song, including ratings and listens.

Properties

- int **TotalListens** `[get, set]`
  Total number of times the song has been rated.
- float **AverageRating** `[get, set]`
- float **MedianRating** `[get, set]`
- int[ ] **RatingDistribution** = new int[20] `[get, set]`
  Distribution of ratings across 20 buckets, grouped by 1-5, 6-10, etc.
- bool **NoData** `[get]`
  True if no ratings of the song exist.

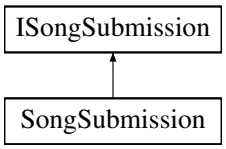
The documentation for this class was generated from the following file:

- MyMusicTaste/Models/Song/SongStats.cs

5.45 SongSubmission Class Reference

Handles submission of new songs to the MongoDB database, ensuring no duplicates exist.

Inheritance diagram for SongSubmission:



Public Member Functions

- async Task< string > SubmitSongAsync (Song song)
  Submits a new song to the database.

5.45.1 Member Function Documentation

5.45.1.1 SubmitSongAsync()

```
async Task< string > SubmitSongAsync (
            Song song)  [inline]
```

Parameters

song	The song to submit.

Returns

A task for the ID of the newly created song as a string.

Exceptions

EntryAlreadyExistsException	Thrown if the song already exists in the repository.

Implements ISongSubmission.

The documentation for this class was generated from the following file:

- MyMusicTaste/Database/Contexts/MongoDb/Operations/MongoSongSubmission.cs

5.46 SongSubmissionPage Class Reference

Page for submitting a new song to the database. Validates user input and handles navigation after submission.

Static Public Member Functions

- static string **GetRoute** ()

Static Public Attributes

- const string **ROUTE_TEMPLATE** = "/submit-song"

Protected Member Functions

- override void **OnInitialized** ()

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-SongSubmission/SongSubmissionPage.razor.cs

5.47 Thumbnail Class Reference

A component that displays an image thumbnail or a placeholder if the image link is invalid.

Protected Member Functions

- override async Task **OnParametersSetAsync** ()

Properties

- string? **ImageLink** `[get, set]`

  The URL of the image to display in the thumbnail.
- string? **AltText** `[get, set]`

  Alternative text to display if the image cannot be shown.
- Dictionary< string, object > **UnmatchedAttributes** = null! `[get, set]`

  Additional HTML attributes to apply to the image or placeholder container.

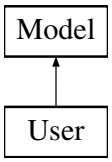
The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Misc/Thumbnail.razor.cs

5.48 User Class Reference

A model representing basic user information.

Inheritance diagram for User:



Properties

- required string **Username** `[get, set]`
- string? **AboutMe** `[get, set]`
- string? **ProfilePictureLink** `[get, set]`
- string? **BannerPictureLink** `[get, set]`

Properties inherited from Model

- ObjectId **Id** `[get, set]`

The documentation for this class was generated from the following file:

- MyMusicTaste/Models/User/User.cs

5.49 UserPage Class Reference

Displays a user's profile including general info, statistics, and comments. Supports editing the profile for the authorized user.

Static Public Member Functions

- static string **GetRoute** (ObjectId userId)

Static Public Attributes

- const string **ROUTE_TEMPLATE** = "/users/{UserId}"

Protected Member Functions

- override async Task **OnInitializedAsync** ()

Properties

- string **UserId** = null! `[get, set]`

  The ID of the user whose profile is being displayed.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-User/UserPage.razor.cs

5.50 UserSearchItem Class Reference

Displays an item in search results that navigates to the user's page when clicked.

Properties

- Models.? User **User** `[get, set]`

  The user data to display.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Search/UserSearchItem.razor.cs

5.51 UserSignupForm Class Reference

A form component that allows new users to sign up for the application. Handles input validation, submission, and automatic login after successful registration.

The documentation for this class was generated from the following file:

- MyMusicTaste/Components/Page-Auth/UserSignupForm.razor.cs

5.52 UserStats Class Reference

Represents statistical information about a user's ratings.

Properties

- List<(string, double)>? **AlbumsByMean** `[get, set]`

  Average ratings per album, as a list of (album name, mean rating) tuples.
- List<(string, double)>? **AuthorsByMean** `[get, set]`

  Average ratings per author, as a list of (author name, mean rating) tuples.
- List<(string, double)>? **GenresByMean** `[get, set]`

  Average ratings per genre, as a list of (genre name, mean rating) tuples.

The documentation for this class was generated from the following file:

- MyMusicTaste/Models/User/UserStats.cs

Index

AssignRoleAsync
  IIdentityProvider, 28
  MongoIdentity, 42
AuthController, 15
  LoginAsync, 15
  LogoutAsync, 16
AuthController.LoginDto, 37
  Password, 38
  Username, 38
AuthorizeUserById
  IIdentityProvider, 29
  MongoIdentity, 42

CalculateSongStatsAsync
  ISongStatsCalculation, 34
  MongoSongsStatsCalculation, 50
CalculateUserStatsAsync
  IUserStatsCalculation, 37
  MongoUserStatsCalculation, 52
Comment, 16
CommentComp, 17
CommentSection, 17
Configure
  MongoIdentity, 43
ConfirmDialog, 18
Connect
  MongoDbContext, 40
CreateAsync
  IDbRepository< TModel >, 26
  MongoRepository< TModel >, 45
CssSize
  MyMusicTaste.Utils, 13

DatabaseOperationException, 19
DeleteAsync
  IDbRepository< TModel >, 26
  MongoRepository< TModel >, 45
Dialog< TResult >, 20
  OpenDialog, 21
DialogBase, 21
DialogResult< TResult >, 22
Dropzone, 22
  SetActive, 23

GetById
  IDbRepository< TModel >, 26
  MongoRepository< TModel >, 46
GetByIdAsync
  IDbRepository< TModel >, 27
  MongoRepository< TModel >, 46
GetByIdsAsync
  IDbRepository< TModel >, 27
  MongoRepository< TModel >, 47
GetCommentsByPageAsync
  ICommentsListing, 24
  MongoCommentsListing, 39
GetCommentsByUserAsync
  ICommentsListing, 24
  MongoCommentsListing, 40
GetRatingsBySongAsync
  ISongRatingListing, 33
  MongoSongRatingListing, 48
GetRatingsByUserAsync
  ISongRatingListing, 33
  MongoSongRatingListing, 48
GetSongRatingAsync
  ISongRatingListing, 34
  MongoSongRatingListing, 49
GetUserId
  IIdentityProvider, 29
  MongoIdentity, 43

Histogram< TValue, TLabel >, 23

ICommentsListing, 24
  GetCommentsByPageAsync, 24
  GetCommentsByUserAsync, 24
IDbRepository< TModel >, 25
  CreateAsync, 26
  DeleteAsync, 26
  GetById, 26
  GetByIdAsync, 27
  GetByIdsAsync, 27
  UpdateAsync, 27
IIdentityProvider, 28
  AssignRoleAsync, 28
  AuthorizeUserById, 29
  GetUserId, 29
  IsAuthenticated, 29
  LoginUserAsync, 29
  LogOutUserAsync, 30
  SignUpUserAsync, 30
InputDialog, 30
IsAuthenticated
  IIdentityProvider, 29
  MongoIdentity, 43
ISearchOperation< TModel >, 32
  SearchAsync, 32
ISongRatingListing, 33
  GetRatingsBySongAsync, 33

GetRatingsByUserAsync, 33
GetSongRatingAsync, 34
ISongStatsCalculation, 34
CalculateSongStatsAsync, 34
ISongSubmission, 35
SubmitSongAsync, 35
IUserLoginDto, 36
IUserSignupDto, 36
IUserStatsCalculation, 37
CalculateUserStatsAsync, 37

LoginAsync
AuthController, 15
LoginPage, 38
LoginUserAsync
IIdentityProvider, 29
MongoIdentity, 43
LogoutAsync
AuthController, 16
LogOutUserAsync
IIdentityProvider, 30
MongoIdentity, 44

Model, 39
MongoCommentsListing, 39
GetCommentsByPageAsync, 39
GetCommentsByUserAsync, 40
MongoDbContext, 40
Connect, 40
MongoIdentity, 41
AssignRoleAsync, 42
AuthorizeUserById, 42
Configure, 43
GetUserId, 43
IsAuthenticated, 43
LoginUserAsync, 43
LogOutUserAsync, 44
MongoIdentity, 42
SignUpUserAsync, 44
MongoRepository< TModel >, 44
CreateAsync, 45
DeleteAsync, 45
GetById, 46
GetByIdAsync, 46
GetByIdsAsync, 47
UpdateAsync, 47
MongoSongRatingListing, 48
GetRatingsBySongAsync, 48
GetRatingsByUserAsync, 48
GetSongRatingAsync, 49
MongoSongSearch, 49
SearchAsync, 49
MongoSongsStatsCalculation, 50
CalculateSongStatsAsync, 50
MongoUserSearch, 51
SearchAsync, 51
MongoUserStatsCalculation, 51
CalculateUserStatsAsync, 52
MyMusicTaste, 9
MyMusicTaste.Components, 9
MyMusicTaste.Components.Comments, 9
MyMusicTaste.Components.Dialogs, 9
MyMusicTaste.Components.Misc, 10
MyMusicTaste.Components.Page_Auth, 10
MyMusicTaste.Components.Page_Home, 10
MyMusicTaste.Components.Page_Search, 10
MyMusicTaste.Components.Page_Song, 10
MyMusicTaste.Components.Page_SongSubmission, 11
MyMusicTaste.Components.Page_User, 11
MyMusicTaste.Database, 11
MyMusicTaste.Database.Contexts, 11
MyMusicTaste.Database.Contexts.MongoDb, 11
MyMusicTaste.Database.Contexts.MongoDb.Operations, 12
MyMusicTaste.Database.Operations, 12
MyMusicTaste.Models, 13
MyMusicTaste.Startup, 13
MyMusicTaste.Utils, 13
CssSize, 13

OpenDialog
Dialog< TResult >, 21

Password
AuthController.LoginDto, 38

RateSongComp, 52

SearchAsync
ISearchOperation< TModel >, 32
MongoSongSearch, 49
MongoUserSearch, 51
SearchComponent< TModel >, 53
SearchPage, 53
SetActive
Dropzone, 23
SignUpUserAsync
IIdentityProvider, 30
MongoIdentity, 44
Song, 54
SongPage, 54
SongRating, 55
SongRatingItem, 56
SongRatingSection, 56
SongSearchItem, 56
SongStats, 57
SongSubmission, 57
SubmitSongAsync, 57
SongSubmissionPage, 58
SubmitSongAsync
ISongSubmission, 35
SongSubmission, 57

Thumbnail, 58

UpdateAsync
IDbRepository< TModel >, 27
MongoRepository< TModel >, 47
User, 59

Username
    AuthController.LoginDto, 38
UserPage, 60
UserSearchItem, 60
UserSignupForm, 61
UserStats, 61